

| Description of Lab Work nº 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Real-Time systems / <i>Sistemas de Tempo Real</i></b>         |
| Year                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2022/2023                                                        |
| Aim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Development of real-time applications using high level languages |
| Classes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5 classes x 3 hours + 12 extra hours (outside)                   |
| Delivery Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>03/12/2022</b>                                                |
| <p>Concrete Objectives:</p> <ol style="list-style-type: none"> <li>1. Java language <ol style="list-style-type: none"> <li>a. Characteristics of this programming language</li> <li>b. The “Java Virtual Machine”(JVM)</li> <li>c. Object-oriented programming</li> <li>d. Java training examples</li> </ol> </li> <li>2. Real-time features of Java <ol style="list-style-type: none"> <li>a. The Thread class and the “Runnable” interface</li> <li>b. Synchronization and communication between Threads.</li> <li>c. Interaction with physical systems; The “Java Native Interface” (JNI)</li> </ol> </li> <li>3. Develop a practical application of Java real-time as described in Annex 1</li> </ol> |                                                                  |

## **Annex 1 – Description of Work**

This work is aimed at providing the concepts of the Java concurrency language features. For such, the development a of real-time solution, for the problem described below, is proposed. The scenario of this problem is related with a real-time control of a manufacturing assembly line that has to split different types of Lego bricks to make certain constructions, as illustrated in Figure 1.



Figure 1– Lego Bricks and construction

The adequate solution for this problem requires the utilization of a set of developing technologies based on Java SDK, Java Native Interface (JNI), and Dynamic-link library (DLL) development, with more emphasis on the Java concurrency features.

For the work development, in the context of classes, it will be used an educational kit simulating a manufacturing distribution conveyor as shown in Figure 2. The connection to the kit is done through the NI USB 6509 Data Acquisition board (DAQ), already used in Labwork 1.

A simulator of the distribution conveyor kit (Figure 3) is also available, which allows continuing the work outside of the lab. This simulator allows testing before running the

implementation in the real kit, which helps eliminate most of the mistakes that might occur during development.

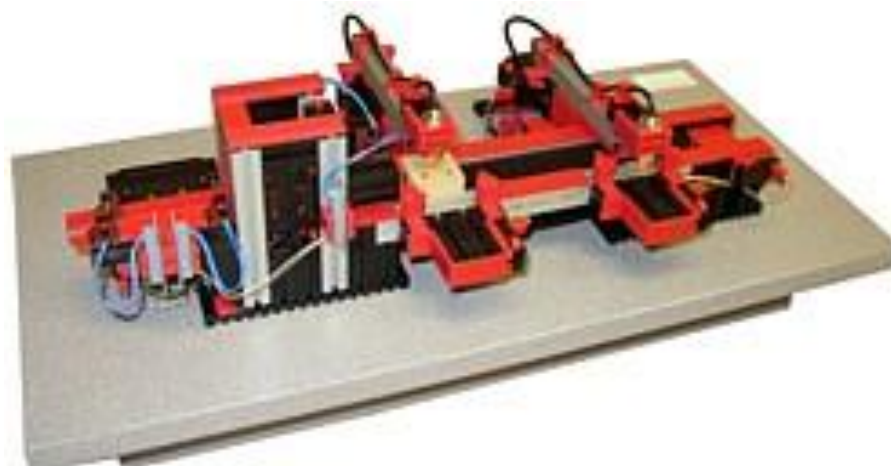


Figure 2– Kit which models the manufacturing assembly line

The mapping of the sensors and actuators of the distribution conveyor into the DAQ is detailed in Figure 3.

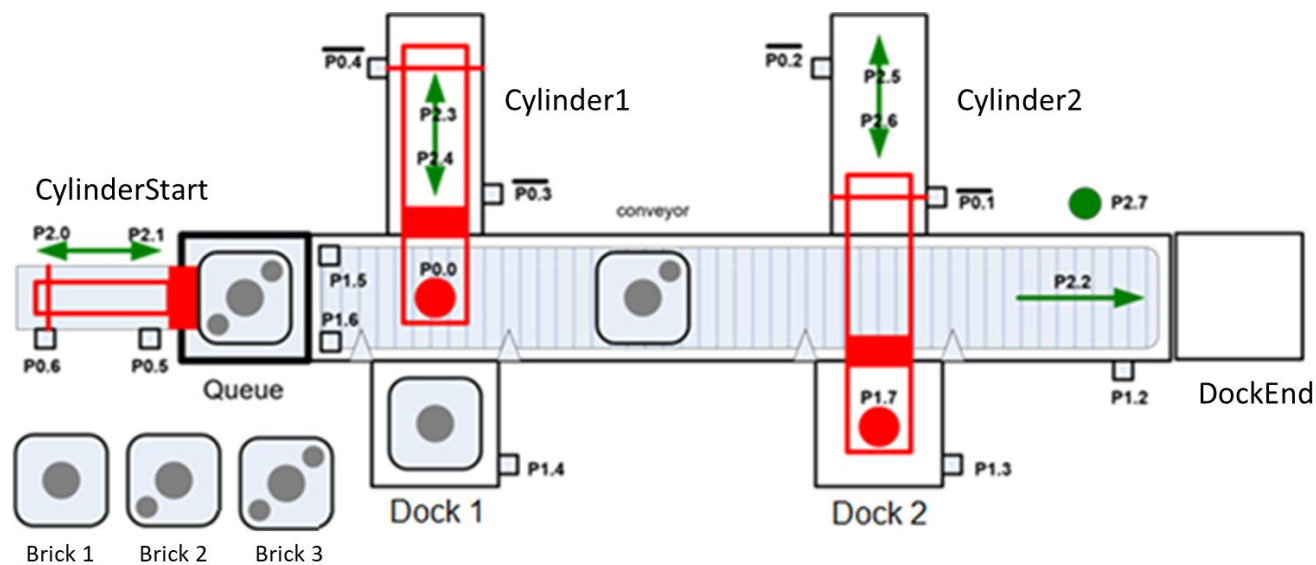


Figure 3 – Sensors and actuators addresses of the distribution line

The manufacturing line has got three distributions docks: Dock 1, Dock 2 and Dock End, which should work in the following way:

- By pressing **“S”** the system **Starts** its working cycle, and by pressing **“F”**, the system **Finishes** its work. (The interaction of the system can be done via console application or through a GUI where the integration of buttons “Start” and “Finish” can be used for this purpose. Please note that the development of a GUI is completely optional and is mostly the responsibility of the student)
- 
- **Dock 1** receives bricks according to the sequence: “Brick1”, “Brick2”, “Brick3”, “Brick1”, “Brick2”, “Brick3”, and so on, letting go any brick which is out of the mentioned sequence. After a complete sequence, the led at P2.7 must flash (On/off) for 1 second.
- **Dock 2** also receives bricks according to the sequence: “Brick1”, “Brick2”, “Brick3”, “Brick1”, “Brick2”, “Brick3”, and so on, letting go any part which is out of the mentioned sequence. After a complete sequence, the led at P2.7 must flash for 2 seconds.
- **Dock End** receives all the bricks that were rejected by dock 1 and dock 2. Whenever this happens, the mentioned led must frantically flash many times for 3 seconds.
- The entry of a new brick in the conveyor is signaled by pressing character “P” on the keyboard (or a corresponding button in a Java UI).
- If a human operator at **dock 1** presses the corresponding button P1.4 one, two or three times, then the next incoming part that must get into dock 1 must be of type “Brick1”, “Brick2”, “Brick3” respectively. Any other type goes to the dock End.
- If a human operator at **dock 2** presses the corresponding button P1.3 one, two or three times, then the next incoming part that must get into dock 2 must be

of type “Brick1”, “Brick2”, “Brick3” respectively. Any other type goes to the dock End.

- If a human operator at **dock End** presses the corresponding button P1.2 one, two or three times, then it means that dock 1, dock 2 or both is/are temporarily (about 5 seconds) closed and the corresponding bricks that should follow be delivered to that docks will proceed to dock end. In this case, the LED should flash for at least 3000 milliseconds to ensure that the operators are aware of the situation.
- It is also necessary to provide an emergency stop and resume operation
- During the emergency stop state, the LED should frantically flash with, e.g. times  $T_{on}=400ms$  and  $T_{off}=400ms$  (but these values can be adjusted).
- The system must remember the total number of bricks of each type that got into each dock, as well as the number of complete sequences (“Brick1”, “Brick2”, “Brick3”).
- Information regarding the state of the system, including statistics, must be shown on the console screen whenever requested (console mode, JAVA Swing, or JavaFX, remember that the user interface is mostly the responsibility of the student).

In terms of operability, each brick is taken from the entry queue, using cylinder start, and placed into the conveyor. There are two sensors at the beginning of the conveyor, which can identify the package type. For instance, a brick of type 3 contains two magnetic marks at their corners, in a diagonal way (see figure 3). Whenever these marks pass under the mentioned sensors, the corresponding brick can be identified.

However, there is a technological constraint on identifying each brick, because when cylinder start completes its onward movement to deliver a brick, the incoming brick is already half inside the conveyor. Therefore, using a non-concurrent approach, in which

the brick is sequentially delivered and then identified, will fail because one of the marks already passed under the sensors, leading to misidentifying the brick. As such, it is recommended that the mentioned identification routine finishes identifying only after each brick arrives to the sensor wired to P0.0 (in Cylinder 1 at dock 1), in which the central mark is identified. Furthermore, a brick of type a only contains the center mark and cannot be spotted by the identification sensors.

Finally, keep in mind that the system has active high and active low sensors. Active low sensors provide logical value false/0 when active, and true/1 otherwise. An example of active low sensor is in P0.4, shown in figure 3. These are usually identified with a top bar. An example of an active high sensor is P0.6 (no top bar).

To develop a solution for adequately coordinating the operation of the distribution conveyor, it is necessary to fulfill the following functional requirements, which are divided into low-level and high-level requirements:

Table 1 – Needed requirements

| Req.                                              | Description                                                                                                                                                                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Low-level requirements (DLL development)</b>   |                                                                                                                                                                                                                               |
| FR_L1                                             | Move and stop each <b>cylinder</b>                                                                                                                                                                                            |
| FR_L2                                             | Move and stop moving the <b>conveyor</b>                                                                                                                                                                                      |
| FR_L3                                             | Read <b>Brick identification</b> and position sensors                                                                                                                                                                         |
| FR_L4                                             | Detect the <b>button/switches</b> states                                                                                                                                                                                      |
| FR_L5                                             | Turn <b>LED</b> on and off                                                                                                                                                                                                    |
| FR_L6                                             | Keyboard or UI interaction                                                                                                                                                                                                    |
| <b>High-level requirements (Java development)</b> |                                                                                                                                                                                                                               |
| FR_H1                                             | Using the keyboard, develop robust (semi-automatic) <b>calibration</b> . Using keys, the user defines the direction of the movements. During a movement, the actuator must stop when it reaches a position sensor (Cylinders) |

|                                           |                                                                                                                                                                                                                                                               |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR_H2                                     | Develop the <b>getBrick()</b> functionality, which gets a brick from the entry queue and puts it in the conveyor                                                                                                                                              |
| FR_H3                                     | By pressing “S” the system Starts its working cycle. Pressing “F”, the system Finishes its work. In a GUI, the Buttons “Start” and “Finish” can be used for this purpose                                                                                      |
| FR_H4                                     | Develop <b>Cylinder_X.gotoPosition(int position)</b> : in which X is related to the corresponding cylinder; position 0 means resting position; 1 means working position (pushing a package)                                                                   |
| FR_H5                                     | Perform <b>identifyBrickType</b> , that identifies each brick with the sensors at the beginning of the conveyor (sensors P1.5 e P1.6); identification ends when brick reaches sensor P0.0 (in cylinder1 at dock 1)                                            |
| FR_H6                                     | Delivering <b>sequence</b> for <b>dock 1</b>                                                                                                                                                                                                                  |
| FR_H7                                     | Delivering <b>sequence</b> for <b>dock 2</b>                                                                                                                                                                                                                  |
| FR_H8                                     | <b>Delivery of bricks</b> out of sequence to <b>dock End</b>                                                                                                                                                                                                  |
| FR_H9                                     | Detection of buttons P1.4 and P1.3 pressed by human operator, at dock 1 and dock 2, and <b>counting the pulses</b> for the corresponding brick                                                                                                                |
| FR_H10                                    | Perform the <b>temporarily close</b> of dock 1 or dock 2 or both according to the detection of button P1.2 pressed with corresponding LED signalization                                                                                                       |
| FR_H11                                    | Handle Emergency <b>STOP/RESUME</b> operation with corresponding LED signalization                                                                                                                                                                            |
| <b>Information retrieval requirements</b> |                                                                                                                                                                                                                                                               |
| FR_H12                                    | <b>Collect statistics</b> of the total number of bricks and sequences that were delivered at each dock. Show these statistics on the console screen whenever requested. If a GUI is used, then these numbers must be updated in the GUI in <b>real-time</b> . |
| <b>Non-Functional requirements:</b>       |                                                                                                                                                                                                                                                               |
| NF1                                       | While operating, the system <b>can accumulate requests</b> .                                                                                                                                                                                                  |
| NF2                                       | While operating, the system can <b>provide information</b> (e.g., number and type of bricks delivered at dock 1)                                                                                                                                              |
| NF3                                       | The system <b>works appropriately</b> (e.g. (i) when placing a new brick on the conveyor, it must be moving; (ii) delivers corresponding bricks to docks according to sequence)                                                                               |



|     |                                                          |
|-----|----------------------------------------------------------|
| NF4 | The system always moves within <b>limits</b> (cylinders) |
|-----|----------------------------------------------------------|

### **Classes plan for the Lab Work 2**

#### **Class 1: [27/28/31 October 2022]**

SW Installation & Interaction with “physical” system. Small Java tutorial and low-level functionalities implementation.

#### **Class 2: [03/04/07 November 2022]**

JNI development for interacting with the physical system from Java. Elaboration of both low-level and high-level routines for LabWork2.

#### **Class 3: [10/11/14 November 2022]**

Introduction to Threads and synchronization and communication mechanisms. Development of the necessary threads for modeling LabWork2 concurrent behavior.

#### **Class 4: [17/18/21 November 2022]**

Fulfillment of each functional requirement towards the solution for LabWork2.

#### **Class 5: [24/25/28 November 2022]**

Conclusion of LabWork2.

#### **Professors:**

- Luis Camarinha-Matos, [cam@uninova.pt](mailto:cam@uninova.pt) (theoretical classes)
- Ana Inês Oliveira, [aio@fct.unl.pt](mailto:aio@fct.unl.pt) (laboratory classes)
- André Rocha, [andre.rocha@uninova.pt](mailto:andre.rocha@uninova.pt) (laboratory classes)
- Filipa Ferrada, [fam@fct.unl.pt](mailto:fam@fct.unl.pt) (laboratory classes)
- Francisco Marques, [fam@uninova.pt](mailto:fam@uninova.pt) (laboratory classes)

#### **Lab Work 2 Delivery:**

Students must submit via Moodle **(and only via Moodle)**:

- ➔ Upload a zip/rar archive with your Visual Studio Project + Netbeans Project.



- ➔ Upload a demo video (aprox. 5/8 minutes) presenting all implemented requirements/functionalities. The video **must be** accompanied by a narration of what is being presented.

### Evaluation criteria

It is provided (as a guiding object) a table with the project evaluation criteria. Each item will be quoted according to the approach and quality of the generated code.

| Items         |                                                                 | Quotation/Penalty |
|---------------|-----------------------------------------------------------------|-------------------|
| Functionality | FR_L1 – Cylinders                                               | 0.25              |
|               | FR_L2 – Conveyor                                                | 0.25              |
|               | FR_L3 – Bricks Identification                                   | 0.25              |
|               | FR_L4 – Switches                                                | 0.25              |
|               | FR_L5 – LED                                                     | 0.25              |
|               | FR_L6 – UI interaction                                          | 0.50              |
|               | FR_H1 –calibration                                              | 0.25              |
|               | FR_H2 –getBrick                                                 | 0.50              |
|               | FR_H3 –Start / finish system                                    | 0.50              |
|               | FR_H4 – cylinder gotoPosition                                   | 0.50              |
|               | FR_H5 –identifyBrickType                                        | 1.00              |
|               | FR_H6 –delivery of sequence for dock 1                          | 1.00              |
|               | FR_H7 – delivery of sequence for dock 2                         | 1.00              |
|               | FR_H8 – delivery of bricks for dock end                         | 1.00              |
|               | FR_H9 –delivery of out of sequence bricks for dock 1 and dock 2 | 1.50              |
|               | FR_H10 –temporarily close dock 1 or dock 2 or both              | 2.00              |
|               | FR_H11 –STOP/RESUME                                             | 1.50              |
|               | FR_H12 –collect statistics                                      | 1.50              |
|               | NF1 – While operating, the system can accumulate requests       | 1.50              |
|               | NF2 – While operating, the system can provide information       | 1.50              |
|               | NF3 – The system works appropriately                            | 0.50              |
|               | NF4 – The system always moves within limits                     | 0.50              |

Departamento de Engenharia Electrotécnica e de Computadores

**Secção de Robótica e Manufatura Integrada**

|                                                                              |              |
|------------------------------------------------------------------------------|--------------|
| Video                                                                        | 2.00         |
| <b>TOTAL</b>                                                                 | <b>20.00</b> |
| No delivery of the VS project or Netbeans project or video (* all MANDATORY) | -20          |